

MANDATORY-HANDS-ON-CTS-DN-5.0

Week 1

EcommercePlatform

EcommerceSearch.java

FactoryMethodPatternExample

Document.class

Document.java

DocumentFactory.class

DocumentFactory.java

ExcelDocument.class

ExcelDocument.java

ExcelDocumentFactory.class

ExcelDocumentFactory.java

FactoryMethodPatternExample.class

FactoryMethodPatternExample.java

PdfDocument.class

PdfDocument.java

PdfDocumentFactory.class

PdfDocumentFactory.java

WordDocument.class

WordDocument.java

WordDocumentFactory.class

WordDocumentFactory.java

FinancialForecasting

src

main

test/java/com/example

AAAPatternTest.java

AssertionsTest.java

MainTest.java

target

OUTLINE

TIMELINE

JAVA PROJECTS

MAVEN

EcommerceSearch.java

Week 1 > EcommercePlatform > EcommerceSearch.java > Product

```
1 class Product {
2     int productId;
3     String productName;
4     String category;
5
6     Product(int productId, String productName, String category) {
7         this.productId = productId;
8         this.productName = productName;
9         this.category = category;
10    }
11
12    void display() {
13        System.out.println(
14            "Product ID: " + productId +
15            ", Name: " + productName +
16            ", Category: " + category
17        );
18    }
19 }
20
21 public class EcommerceSearch {
22
23     // Linear Search using productId
24     public static Product linearSearch(Product[] products, int searchId) {
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SQL HISTORY

TASK MONITOR

zsh

zsh Week 1

zsh Week 1

Run: Ecom...

mathesh@Matheshs-MacBook-Air Mandatory-Hands-on-CTS-DN-5.0 % /usr/bin/env /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/mathesh/Library/Application\ Support/Code/User/workspaceStorage/0b7345b8e7d472645cddd170427bbea1/redhat.java/jdt_ws/jdt.ls-java-project/bin EcommerceSearch

Linear Search Result:

Product ID: 104, Name: Bag, Category: Fashion

Binary Search Result:

Product ID: 104, Name: Bag, Category: Fashion

mathesh@Matheshs-MacBook-Air Mandatory-Hands-on-CTS-DN-5.0 %

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

EcommerceSearch.java

Describe what to build

Local

Default Approvals

Ln 1, Col 1 Spaces: 4 UTF-8 LF () Java

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FinancialForecasting

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junit_setup_examples

.vscode

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OUTLINE

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JAVA PROJECTS

MAVEN

FinancialForecasting.java 1 X

Week 1 > FinancialForecasting > FinancialForecasting.java > FinancialForecasting

```
1 public class FinancialForecasting {
2
3     // Recursive method to calculate future value
4     public static double calculateFutureValue(double currentValue, double growthRate, int years) {
5
6         // Base case
7         if (years == 0) {
8             return currentValue;
9         }
10
11        // Recursive case
12        return calculateFutureValue(currentValue, growthRate, years - 1)
13            * (1 + growthRate);
14    }
15
16    Run | Debug
17    public static void main(String[] args) {
18
19        double currentValue = 10000;
20        double growthRate = 0.08; // 8% growth rate
21        int years = 5;
22
23        double futureValue = calculateFutureValue(currentValue, growthRate, years);
24        System.out.println("Current Value: Rs. " + currentValue);
25    }
26 }
```

PROBLEMS 1

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TASK MONITOR

```
mathesh@Matheshs-MacBook-Air Mandatory-Hands-on-CTS-DN-5.0 % /usr/bin/env /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/mathesh/Library/Application\ Support/Code/User/workspaceStorage/0b7345b8e7d472645cddd170427bbea1/redhat.java/jdt_ws/jdt.ls-java-project/bin FinancialForecasting
Current Value: Rs. 10000.0
Growth Rate: 8.0%
Number of Years: 5
Predicted Future Value: Rs. 14693.28
mathesh@Matheshs-MacBook-Air Mandatory-Hands-on-CTS-DN-5.0 %
```

zsh

zsh Week 1

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Run: Ecom...

Run: Finan...

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+ FinancialForecasting.java

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+ | | Auto |

Local | Default Approvals

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